A Project Report on

NirmalSagar: Underwater Marine Litter & Plastic Detection

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Computer Science & Engineering (Data Science)

by

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Marine debris such as plastic waste, fishing nets, bottles, and other submerged objects is a major concern for marine life and underwater ecosystems. Detection of such debris through manual monitoring is time-consuming and requires significant human effort. Moreover, underwater conditions such as turbidity, low light, poor visibility, lighting variations, and complex backgrounds make automated debris detection challenging. The project proposes an AI-based computer vision system using deep learning for the automated detection and localization of submerged marine debris from underwater images. The proposed detection pipeline combines underwater image enhancement and YOLO-based object detection for identifying marine litter in challenging underwater scenes. The system also focuses on domain adaptation and generalisation strategies, including synthetic augmentation, style transfer, and adversarial domain adaptation, to improve the ability of models trained on datasets such as TrashCan 1.0 and JAMSTEC to generalise to unseen and murkier Indian coastal and seafloor conditions. The project further aims to benchmark the model across multiple heterogeneous underwater marine-debris datasets using standard evaluation metrics such as mAP, IoU, and F1 score. A debris-density estimation module is also proposed to aggregate per-frame detections into an area-level pollution score, supporting marine pollution monitoring and cleanup prioritisation along Indian coastlines. The project uses Python, Deep Learning, Computer Vision, OpenCV, image processing, and YOLOv11 for developing and adapting the detection pipeline.

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Chapter 1

Introduction

Marine pollution has become an important environmental concern due to the increasing amount of waste entering oceans, coastal regions, and underwater environments. Marine debris includes plastics, fishing nets, bottles, containers, and other forms of waste that can remain in aquatic environments for long periods. Such debris can affect marine ecosystems and may pose risks to marine organisms through ingestion, entanglement, and habitat disturbance. Monitoring and identifying marine debris is therefore important for understanding underwater pollution and supporting effective cleanup and conservation activities.

Underwater marine debris detection is a challenging computer vision problem because underwater images often differ significantly from images captured in normal terrestrial environments. Factors such as turbidity, low visibility, low light, lighting variations, and complex underwater backgrounds can make debris difficult to distinguish from surrounding objects. Changes in water conditions can also affect the appearance, colour, contrast, and visibility of marine litter. As a result, conventional image-based detection methods may experience difficulties when applied to underwater environments with different visual characteristics.

Manual inspection of underwater environments is another major challenge. Identifying and recording marine debris through human inspection requires considerable time and effort, particularly when large underwater areas have to be monitored. The process can also become difficult when visibility is poor or when debris is partially hidden within complex underwater backgrounds. These limitations create a need for automated computer vision and artificial intelligence techniques that can assist in detecting and localizing marine debris from underwater images.

The proposed project, NirmalSagar: Underwater Marine Litter & Plastic Detection, focuses on developing an AI-based approach for automated detection and localization of underwater marine debris. The system combines image enhancement and YOLO-based object detection to improve the identification of marine litter under challenging underwater conditions. The project also considers the problem of domain shift, where a model trained using one underwater dataset or environment may not perform equally well when applied to another environment with different visual characteristics.

To address this challenge, the project focuses on domain adaptation and generalisation strategies using techniques such as synthetic augmentation, style transfer, and adversarial domain adaptation. These approaches are intended to improve the ability of the trained model to generalise from existing datasets such as TrashCan and JAMSTEC to unseen and

murkier Indian coastal and seafloor conditions. The performance of the detection system is evaluated across heterogeneous underwater marine-debris datasets using standard evaluation measures such as mAP, IoU, and F1-score.

In addition to object detection, the project considers the estimation of debris density from detected objects. Per-frame detections can be aggregated to obtain an area-level pollution score, which can provide useful information about the concentration of marine debris in a particular area. Such an approach can support the prioritisation of cleanup activities along Indian coastlines. Therefore, the overall project focuses not only on detecting individual marine debris objects but also on developing a broader approach for automated underwater pollution monitoring.

1.1 Motivation

The motivation for this project comes from the increasing need for efficient monitoring of marine pollution and the limitations associated with manual underwater inspection. Marine debris such as plastics, fishing nets, bottles, and other waste can remain in underwater environments and may affect marine ecosystems and marine life. Identifying such debris at an early stage can support better monitoring and help in planning appropriate cleanup activities.

Underwater environments present several visual challenges for automated detection. Turbidity can reduce visibility, while low light and lighting variations can change the appearance of objects in underwater images. Complex backgrounds can further make it difficult to distinguish marine debris from natural underwater objects. These challenges make underwater marine debris detection different from conventional object detection performed on clear images.

Another motivation is the time and human effort required for manual monitoring. When underwater environments need to be inspected over large areas or across a large number of images, manually identifying individual debris objects becomes difficult and time-consuming. An automated AI-based system can assist in processing underwater images and identifying possible debris objects more efficiently.

The project is also motivated by the need for models that can work beyond the conditions represented in their training datasets. A model trained on a particular dataset may encounter difficulties when applied to an unseen underwater environment having different levels of turbidity, lighting, background complexity, or visual appearance. Therefore, domain adaptation and generalisation are important considerations for developing a detection system that can operate under different underwater conditions.

The proposed NirmalSagar system aims to address these challenges by combining image enhancement, deep learning, computer vision, and YOLO-based object detection. The project further investigates generalisation across multiple heterogeneous datasets and considers murkier Indian coastal and seafloor conditions. This can contribute towards faster and more consistent marine pollution monitoring while reducing dependence on completely manual inspection.

1.2 Problem Statement

The detection of marine debris in underwater environments is difficult because underwater images are affected by several environmental and visual factors. Turbidity, low visibility, low light, lighting variations, and complex backgrounds can significantly change the appearance of objects in an image. Marine debris may also be partially obscured or difficult to distinguish from surrounding underwater structures.

Marine litter such as plastics, fishing nets, bottles, and other waste can pose a threat to marine life and underwater ecosystems. However, identifying these objects manually from underwater images requires significant human effort and can become time-consuming when a large number of images or large underwater areas need to be inspected. This creates a requirement for automated techniques that can assist in identifying and localizing marine debris efficiently.

A further challenge is the variation between different underwater environments and datasets. A detection model trained using images collected under one set of environmental conditions may not provide the same level of performance when applied to images from another location. Differences in water clarity, lighting, background characteristics, and object appearance can result in domain-shift degradation.

Therefore, the problem addressed by this project is the development of an AI-based system for automatic detection and localization of submerged marine debris from underwater images while considering challenging underwater conditions and differences between datasets. The project focuses on combining image enhancement with YOLO-based detection and evaluating domain adaptation and generalisation strategies to improve performance across heterogeneous underwater environments.

1.3 Objectives

The objectives of the proposed NirmalSagar project are derived from the challenges associated with underwater marine debris detection, dataset variation, and marine pollution monitoring.

- To develop an underwater debris detection pipeline that combines image enhancement with YOLO for accurate detection and classification of marine litter from underwater images.
- To investigate the effect of challenging underwater conditions such as turbidity, low visibility, low light, lighting variations, and complex backgrounds on marine debris detection.
- To develop and evaluate domain adaptation and generalisation strategies including synthetic augmentation, style transfer, and adversarial domain adaptation so that a model trained on one dataset or site, such as TrashCan and JAMSTEC, can generalise to unseen and murkier Indian coastal and seafloor conditions.

- To benchmark the performance of the detection system across multiple heterogeneous datasets including TrashCan 1.0, Underwater Marine Debris in Shallow Waters, and JAMSTEC Deep-sea Debris Database.
- To quantify domain-shift degradation using standard evaluation metrics such as mean Average Precision (mAP), Intersection over Union (IoU), and F1-score.
- To develop a debris-density estimation approach that aggregates per-frame detections into an area-level pollution score for representing the concentration of detected marine debris.
- To support the prioritisation of cleanup activities along Indian coastlines by providing an area-level indication of marine debris density.

1.4 Scope

The scope of the NirmalSagar project covers the automated analysis of underwater images for detecting and localizing marine debris. The project focuses on computer vision and deep learning techniques that can assist in identifying marine litter under challenging underwater conditions.

- **Automated Marine Debris Detection:**

  The system focuses on automated detection and localization of submerged marine debris from underwater images using deep learning and computer vision techniques.

- **Underwater Image Analysis:**

  The project considers underwater images affected by challenging conditions such as low visibility, turbidity, low light, lighting variations, and complex backgrounds.

- **Image Enhancement:**

  Image enhancement techniques are considered as part of the detection pipeline to improve the visual characteristics of underwater images before applying object detection.

- **YOLO-Based Object Detection:**

  The project uses YOLO-based object detection for identifying and localizing marine litter within underwater images.

- **Domain Adaptation and Generalisation:**

  The project considers synthetic augmentation, style transfer, and adversarial domain adaptation strategies to improve model generalisation across different underwater environments and datasets.

- **Multi-Dataset Evaluation:**

  The detection system is evaluated across heterogeneous marine-debris datasets including TrashCan 1.0, Underwater Marine Debris in Shallow Waters, and JAMSTEC Deep-sea Debris Database.

- **Performance Evaluation:**

  The project evaluates detection performance using standard metrics such as mAP, IoU, and F1-score, while also considering the degradation caused by domain shifts between datasets and environments.

- **Debris-Density Estimation:**

  The project includes aggregation of per-frame detections to estimate an area-level pollution score representing debris density.

- **Indian Coastal and Seafloor Conditions:**

  The project specifically considers the generalisation of detection models towards unseen and murkier Indian coastal and seafloor conditions.

- **Marine Pollution Monitoring:**

  The resulting debris-density information can support marine pollution monitoring and help in prioritising cleanup activities along Indian coastlines.

Chapter 2

Literature Review

In recent years, there has been growing interest in the application of artificial intelligence, deep learning, and computer vision for underwater object and marine debris detection. Marine debris detection is challenging because underwater images are affected by factors such as turbidity, low visibility, lighting variations, complex backgrounds, and the presence of small or camouflaged objects. Several studies have explored different object detection architectures, underwater marine-debris datasets, image enhancement techniques, and domain generalisation approaches. This literature review summarizes important developments in underwater object and marine-debris detection and identifies the research gaps that motivate the proposed NirmalSagar system.

2.1 Comparative Analysis of Recent Studies

Different approaches for underwater object and marine-debris detection have been reviewed in recent studies. Traditional hand-crafted features as well as deep learning-based object detection architectures have been explored for identifying underwater objects. Commonly investigated architectures include SSD, YOLO-based models, FPN, PANet, BiFPN, Faster R-CNN, and other deep learning approaches. Although these methods have improved automated detection, challenges related to small objects, camouflage, turbidity, dataset imbalance, and generalisation across underwater environments remain.

The development of deep learning-based marine debris detection has provided improved capabilities for automated detection and monitoring. Survey studies have highlighted the increasing use of deep learning architectures for underwater object detection and have also identified limitations in the diversity and generalisability of available datasets. Existing datasets and studies differ in terms of underwater environments, camera conditions, object categories, and image quality, making it difficult to achieve reliable performance across different locations.

Several studies have specifically focused on marine-debris datasets and detection models. Duraš et al. [5] introduced the SeaClear multi-domain dataset containing 8,610 images and 40 classes and evaluated baseline detection using Faster R-CNN and YOLOv6. The study also investigated image enhancement and highlighted challenges related to imbalanced categories and cross-site and cross-camera generalisation.

Jiang et al. [6] proposed an improved YOLOv8m model using a C2f-CloFormer backbone, C2f-SCConv neck modules, and WIoU v3 regression loss for marine-debris identification and classification. Although the approach improves the detection model, the study reports a trade-off between accuracy and inference speed, with lower inference FPS than the baseline. Recognition performance also decreases for some debris categories.

Zhang et al. [7] proposed an improved YOLOv11 model using a FasterNet backbone with C2f-Faster and SE channel-attention modules for lightweight underwater marine-debris detection. The study focuses on improving detection efficiency, although large-scale variation of underwater targets and near/far distortion remain unresolved challenges.

Khriss et al. [8] comparatively evaluated Faster R-CNN, SSD, YOLOv8, and YOLOv9 for underwater plastic-debris detection and monitoring. The study highlights challenges in detecting rare or less-common objects and reports that cluttered and noisy underwater scenes continue to affect detection performance.

Zhang et al. [9] proposed UDD-YOLO, which enhances a YOLOv12n baseline using a Cold Diffusion preprocessor, an AMC2f multi-scale feature module, and a Unified-IoU loss. The approach addresses underwater image quality and marine-debris detection; however, the diffusion preprocessing introduces additional processing latency on edge hardware.

The reviewed survey studies also provide important observations regarding the broader research area. Er et al. [1] reviewed deep learning architectures and popular datasets for underwater marine object detection and highlighted the absence of a universally accepted benchmark dataset, inter-class similarity, camouflage, and dataset imbalance. Jian et al. [2] surveyed traditional and deep learning-based underwater object detection methods and identified weaknesses in detecting small and camouflaged targets along with limitations in dataset diversity and generalisability.

Moorton et al. [3] reviewed deep learning applications for tracking and detecting marine debris and conducted an original YOLOv5 experiment using a small custom dataset. The study highlighted limitations associated with small datasets, limited public availability of data, and the exclusion of micro- and nano-debris. Moorton et al. [4] further conducted a PRISMA-ScR scoping review of object detection methods for macro marine debris and identified limited diversity in benchmark datasets and inconsistent reporting of metrics and data.

2.2 Research Gap

The literature review highlights that although considerable progress has been made in applying deep learning to underwater object and marine-debris detection, several limitations remain. Existing annotated underwater marine-debris datasets have limited diversity, while variations in underwater environments, cameras, and sites make generalisation difficult. Poor image quality caused by turbidity, low light, and colour distortion further affects the visibility of marine debris. Small, rare, and underrepresented debris categories also remain challenging because of class imbalance and the difficulty of detecting small or camouflaged objects.

Another important limitation is the limited generalisation of models across different un-

derwater environments. Models trained on a particular dataset or site may not perform consistently when applied to unseen and murkier underwater conditions. Differences in image quality, environmental conditions, camera characteristics, and debris categories contribute to domain-shift degradation.

Therefore, the proposed work focuses on using multiple publicly available underwater marine-debris datasets to provide greater variation in debris types and underwater conditions. Underwater image enhancement and preprocessing are considered to improve the visibility of debris features before detection. Training and evaluation on diverse underwater imagery are intended to improve robustness beyond a single dataset or environment. YOLO-based detection with suitable data augmentation and class-balancing techniques is considered to improve the detection of small, difficult, and underrepresented debris categories.

The research gap identified from the reviewed studies therefore motivates the development of an AI-based marine-debris detection system that combines underwater image processing, deep learning-based object detection, and evaluation across diverse underwater datasets. The proposed NirmalSagar system is aimed at addressing the challenges of automated marine-debris detection and localisation under difficult underwater conditions.

Ref.	**Author(s) and Year**	**Key Contributions**	**Limitations / Gaps Identified**
[1]	Er, M.J., Chen, J., Zhang, Y., Gao, W. (2023)	Reviewed deep learning-based underwater marine object detection architectures including SSD, YOLOv3, FPN, PANet, BiFPN, and Faster R-CNN and organized the research challenges into four categories.	No universally accepted benchmark dataset exists. Inter-class similarity and camouflage remain unresolved, while reviewed datasets show imbalance and small-scale objects.
[2]	Jian, M., Yang, N., Tao, C., Zhi, H., Luo, H. (2024)	Surveyed 14 traditional hand-crafted-feature methods and 34 deep learning methods across eight categories and reviewed representative underwater object-detection datasets.	Deep learning models remain weak on small and camouflaged targets. Turbidity and refraction remain challenging, while existing datasets lack diversity and generalisability.
[3]	Moorton, Z., Kurt, Z., Woo, W.L. (2024)	Reviewed 28 deep learning studies on marine-debris detection and conducted an original YOLOv5 experiment using a small custom litter-versus-animal dataset.	The custom dataset was too small for reliable results, with maximum reported accuracy of about 63%. Most surveyed studies did not share data publicly and micro- and nano-debris were excluded.
[4]	Moorton, Z., Mistry, K., Strachan, R., Hu, S. (2025)	Conducted a PRISMA-ScR scoping review of 25 papers with comparative analysis of models, datasets, and hardware and proposed a reporting-standardisation framework.	Benchmark datasets lack diversity and are largely based on JAMSTEC-derived deep-sea imagery. Inconsistent metric and data reporting limits comparison between studies.

Ref.	Author(s) and Year	Key Contributions	Limitations / Gaps Identified
[5]	Đuraš, A., Wolf, B.J., Ilioudi, A., Palunko, I., De Schutter, B. (2024)	Introduced the SeaClear multi-domain dataset containing 8,610 images and 40 classes and evaluated baseline detection using Faster R-CNN and YOLOv6 along with image-enhancement analysis.	Category distribution is imbalanced across sites and cameras. Image enhancement did not improve cross-site and cross-camera generalisation, and some sites contain very few images.
[6]	Jiang, W., Yang, L., Bu, Y. (2024)	Proposed an improved YOLOv8m using a C2f-CloFormer backbone, C2f-SCConv neck modules, and WIoU v3 regression loss for marine-debris identification and classification.	Lower inference FPS than the baseline creates an accuracy-speed trade-off. The model is not fully lightweight and recognition decreases for some classes such as can, branch, wreckage, and tarp.
[7]	Zhang, L., Yang, J., Yu, H. (2025)	Improved YOLOv11 using a FasterNet backbone with C2f-Faster and SE channel-attention modules for lightweight underwater marine-debris detection.	No explicit limitations section was reported. Large-scale variation of underwater targets and near/far distortion remain unresolved challenges.
[8]	Khriss, A., Kerkour Elmiad, A., Badaoui, M., Barkaoui, A.-E., Zarhloule, Y. (2024)	Compared Faster R-CNN, SSD, YOLOv8, and YOLOv9 for underwater plastic-debris detection and monitoring.	Poor detection of rare or less-common objects and varying model stability remain challenges. Cluttered and noisy underwater scenes also affect detection.
[9]	Zhang, C., Zhang, W., Liang, C., Qiao, Q., Tao, J. (2025)	Proposed UDD-YOLO using a YOLOv12n baseline enhanced with a Cold Diffusion preprocessor, AMC2f multi-scale feature module, and Unified-IoU loss.	Diffusion preprocessing introduces additional per-frame latency on edge hardware, with the reported FPS decreasing from 79.1 to 65.9 on Jetson Xavier NX.

Table 2.1: Summary of Literature Review on Underwater Marine Debris Detection

Chapter 3

Project Design

3.1 Existing System

The existing approaches for underwater marine debris monitoring mainly depend on manual inspection and image-based detection methods developed for specific underwater environments. Manual inspection involves examining underwater environments and identifying marine debris such as plastics, fishing nets, bottles, and other waste through human observation. Although manual monitoring can provide useful information about underwater pollution, it requires considerable time and human effort, especially when large underwater areas or a large number of images need to be inspected.

Underwater image-based detection systems have also been developed using computer vision and deep learning techniques. However, the performance of such systems can be affected by the visual characteristics of underwater environments. Factors such as turbidity, low visibility, low light, lighting variations, and complex backgrounds can make marine debris difficult to identify. Objects may appear different depending on water conditions, illumination, and the surrounding underwater environment.

Another limitation of existing detection approaches is their dependency on the dataset and environment used during training. A model trained on images collected from one location or dataset may not provide the same performance when applied to an unseen environment. Differences in water clarity, lighting conditions, background characteristics, and object appearance can result in domain-shift degradation.

Existing approaches may therefore perform well under the conditions represented in their training datasets but may face difficulties when applied to different or murkier underwater environments. This is particularly important for marine debris monitoring in Indian coastal and seafloor conditions, where underwater visual characteristics may differ from those present in existing benchmark datasets.

Thus, the existing system can be summarized as follows:

- Manual underwater inspection requires significant time and human effort for identifying marine debris.
- Underwater image quality is affected by turbidity, low visibility, low light, lighting variations, and complex backgrounds.
- Conventional detection approaches may have difficulty distinguishing marine debris from

surrounding underwater objects.

- Models trained on a particular dataset or environment may experience performance degradation when applied to unseen underwater conditions.
- Existing approaches require improved generalisation strategies to handle differences between heterogeneous underwater datasets and real-world environments.

3.2 Proposed System

The proposed system, NirmalSagar: Underwater Marine Litter & Plastic Detection, is designed as an AI-based approach for automated detection and localization of marine debris from underwater images. The system focuses on addressing the challenges associated with underwater visual conditions and differences between datasets.

The proposed detection pipeline combines underwater image enhancement with YOLO-based object detection. Image enhancement is used as part of the processing pipeline to improve the visual characteristics of underwater images affected by conditions such as low visibility, turbidity, low light, and lighting variations. The enhanced images are then processed using a YOLO-based detection approach to identify and localize marine debris.

A major component of the proposed system is the consideration of domain adaptation and generalisation. A model trained using one underwater dataset or site may not perform equally well when applied to another environment. To address this issue, the project considers strategies including synthetic augmentation, style transfer, and adversarial domain adaptation. These approaches are intended to improve the ability of the detection model to generalise from existing datasets such as TrashCan and JAMSTEC to unseen and murkier Indian coastal and seafloor conditions.

The proposed system also evaluates the detection approach across multiple heterogeneous marine-debris datasets. The datasets considered for benchmarking include TrashCan 1.0, Underwater Marine Debris in Shallow Waters, and the JAMSTEC Deep-sea Debris Database. Performance is evaluated using standard metrics such as mean Average Precision (mAP), Intersection over Union (IoU), and F1-score. These measures help quantify the effectiveness of the detection system and the degradation caused by domain shifts.

In addition to individual object detection, the proposed system includes a debris-density estimation approach. Per-frame detections can be aggregated to estimate an area-level pollution score representing the concentration of marine debris in a particular area. This information can support marine pollution monitoring and help in prioritising cleanup activities along Indian coastlines.

The overall proposed system therefore focuses on combining image enhancement, deep learning, YOLO-based object detection, domain adaptation and generalisation, multi-dataset evaluation, and debris-density estimation into a unified approach for underwater marine debris monitoring.

The major components of the proposed system are:

- **Underwater Image Input:** Underwater images are provided as input to the detection pipeline for analysis of marine debris.
- **Image Enhancement:** Image enhancement is applied to improve the visual characteristics of underwater images affected by challenging environmental conditions.
- **YOLO-Based Detection:** A YOLO-based object detection approach is used to identify and localize marine debris within underwater images.

- **Domain Adaptation and Generalisation:** Synthetic augmentation, style transfer, and adversarial domain adaptation strategies are considered to improve model generalisation across different underwater environments.
- **Multi-Dataset Benchmarking:** The detection system is evaluated using TrashCan 1.0, Underwater Marine Debris in Shallow Waters, and the JAMSTEC Deep-sea Debris Database.
- **Performance Evaluation:** The system is evaluated using mAP, IoU, and F1-score to measure detection performance and quantify domain-shift degradation.
- **Debris-Density Estimation:** Per-frame detections are aggregated to estimate an area-level pollution score representing the density of marine debris.
- **Marine Pollution Monitoring:** The estimated debris density can support the prioritisation of cleanup activities along Indian coastlines.

3.3 Critical Components of System Architecture

The proposed NirmalSagar system consists of a sequence of interconnected components that together form the complete underwater marine debris detection pipeline. The architecture begins with the acquisition of underwater images or video sequences and proceeds through data preparation, YOLOv11 model training, inference, and generation of detection results. The major components of the proposed architecture are described below.

3.3.1 Data Acquisition

The first stage of the proposed system is data acquisition. Underwater images and video sequences are collected from underwater environments using suitable underwater cameras and ROV/AUV-based systems. The collected data can represent different underwater conditions, including variations in lighting, turbidity, and depth.

The acquired images and video sequences provide the input data required for subsequent preprocessing, annotation, model training, and evaluation. A diverse collection of underwater data is important because marine debris can appear differently under different environmental conditions.

3.3.2 Data Preparation

The collected underwater data undergoes several preparation steps before being provided to the detection model. The data preparation stage consists of preprocessing, annotation, data augmentation, and dataset splitting.

Preprocessing: The underwater images are prepared by performing operations such as resizing, denoising, and colour and contrast enhancement. These operations help improve the visual characteristics of the images before they are used by the detection model.

Annotation: The marine debris present in the images is annotated using bounding boxes and corresponding class labels. These annotations provide the information required by the object detection model to learn the location and category of marine debris.

Data Augmentation: Data augmentation is applied to increase the diversity of the training data. The architecture considers augmentation operations such as flipping, rotation, Mosaic,

MixUp, and colour jitter. These transformations help expose the model to variations in the appearance and orientation of underwater objects.

Dataset Split: The prepared dataset is divided into training, validation, and testing data. Training data is used for learning the model parameters, validation data is used during model development, and test data is used to evaluate the trained model on previously unseen data.

3.3.3 YOLOv11 Model Training

The prepared training data is provided to the YOLOv11 model during the training stage. The YOLOv11 architecture consists of three major components: the backbone, neck, and detection head.

Backbone: The backbone performs feature extraction from the input images. It learns visual features that help distinguish marine debris from the surrounding underwater environment.

Neck: The neck performs feature fusion by combining information from different feature levels. This helps the detection model use both detailed and higher-level visual information during object detection.

Detection Head: The detection head produces the final detection-related outputs. These include class probabilities, bounding box regression, and objectness scores for the detected objects.

During training, the model uses a training strategy involving loss calculation, optimization, learning-rate scheduling, hyperparameter tuning, early stopping, and model checkpointing. The model parameters are updated through backpropagation and weight updates based on the training process.

3.3.4 Trained Model

After completion of the training process, the learned YOLOv11 model is saved as the trained model. The trained model contains the learned parameters required to detect marine debris in new underwater images or video sequences.

The trained model can subsequently be used during the inference stage to process previously unseen underwater data. The separation between training and inference allows the trained model to be evaluated independently on test data and new underwater inputs.

3.3.5 Inference and Detection

The inference stage uses the trained YOLOv11 model to detect marine debris in new underwater images or video sequences. The input undergoes preprocessing similar to the training pipeline, including resizing, enhancement, and normalization.

The preprocessed input is then passed through the YOLOv11 model during the forward pass. Non-Maximum Suppression (NMS) is applied to remove overlapping detections and retain the relevant bounding boxes.

The detection stage produces information including the class label, bounding box coordinates, and confidence score for each detected marine debris object. These results provide the basis for further analysis and visualization.

3.3.6 Output and Applications

The final stage of the architecture produces annotated detection results. Detected marine debris objects are represented using bounding boxes along with their corresponding class labels and confidence scores.

The resulting information can be used for underwater pollution monitoring and marine conservation activities. The detected data can also support data analysis and reporting and provide information that can assist authorities in making decisions related to marine pollution monitoring and cleanup activities.

The complete architecture therefore establishes a flow from underwater data acquisition to data preparation, YOLOv11 training, model generation, inference, and final detection output. This integrated pipeline provides the foundation for automated underwater marine debris detection using deep learning.

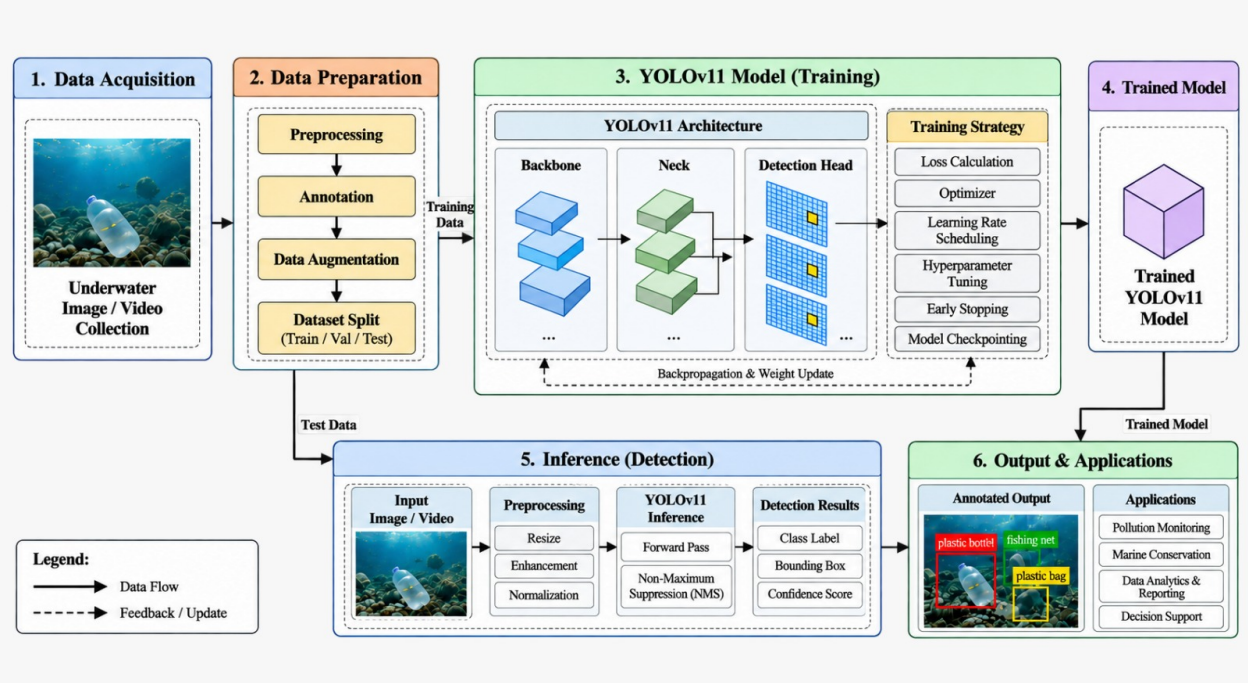


Figure 3.1: Overall Architecture of the Proposed Marine Debris Detection System using YOLOv11

3.4 System Diagrams

System diagrams provide a visual representation of the structure, behaviour, and interaction of the proposed NirmalSagar system. They help in understanding the different components of the system and the flow of information between the user and the detection modules. The major diagrams used for the proposed system are the UML Diagram, Activity Diagram, Use Case Diagram, and Sequence Diagram.

3.4.1 UML Diagram

The UML diagram represents the major classes and components involved in the NirmalSagar system. It shows the structural relationship between the user, preprocessing module, dataset, YOLOv11 model, detection result, image enhancement, and debris analysis components.

The *User* interacts with the system by providing underwater images or videos and viewing the generated detection results. The *Preprocessor* performs the required preprocessing operations before the data is passed to the YOLOv11 model. The *Dataset* represents the collection of underwater marine debris images used for training and evaluation.

The *YOLOv11 Model* performs marine debris detection on the processed input. The detection results are then passed to the *Debris Analyzer*, which can be used to analyse the detected debris and estimate pollution-related information. The *Detection Result* stores the output generated by the detection process. The relationships between these classes describe the flow of data through the proposed system.

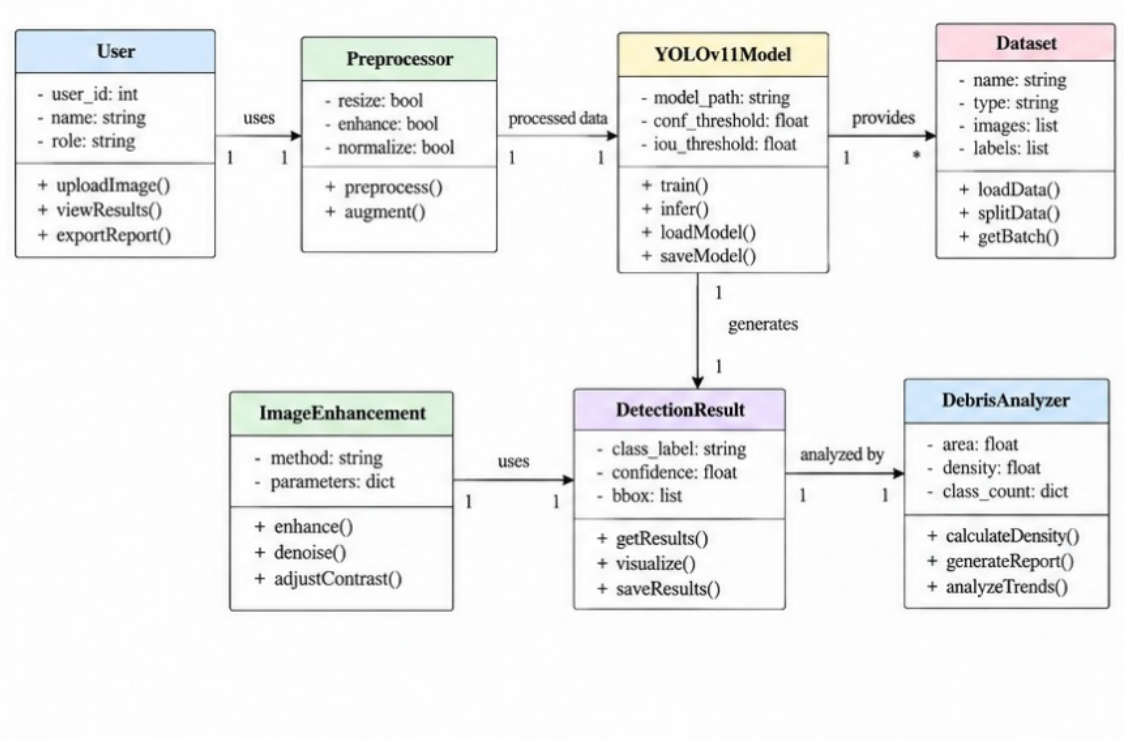


Figure 3.2: UML Class Diagram of the Proposed NirmalSagar System

3.4.2 Activity Diagram

The activity diagram represents the workflow of the NirmalSagar system from input acquisition to the generation of final detection results. The process begins when an underwater image or video is provided to the system. The input is then passed through preprocessing before further processing.

The diagram represents two major workflows: model training and model inference. During training, images are annotated, augmented, divided into appropriate dataset splits, and used to train the YOLOv11 model. The trained model is evaluated and saved for future inference.

During inference, the trained model is loaded and a new underwater image is preprocessed.

YOLOv11 performs detection followed by Non-Maximum Suppression (NMS) to remove overlapping detections. The detection results are then generated, debris density can be estimated, and the final results are displayed or stored.

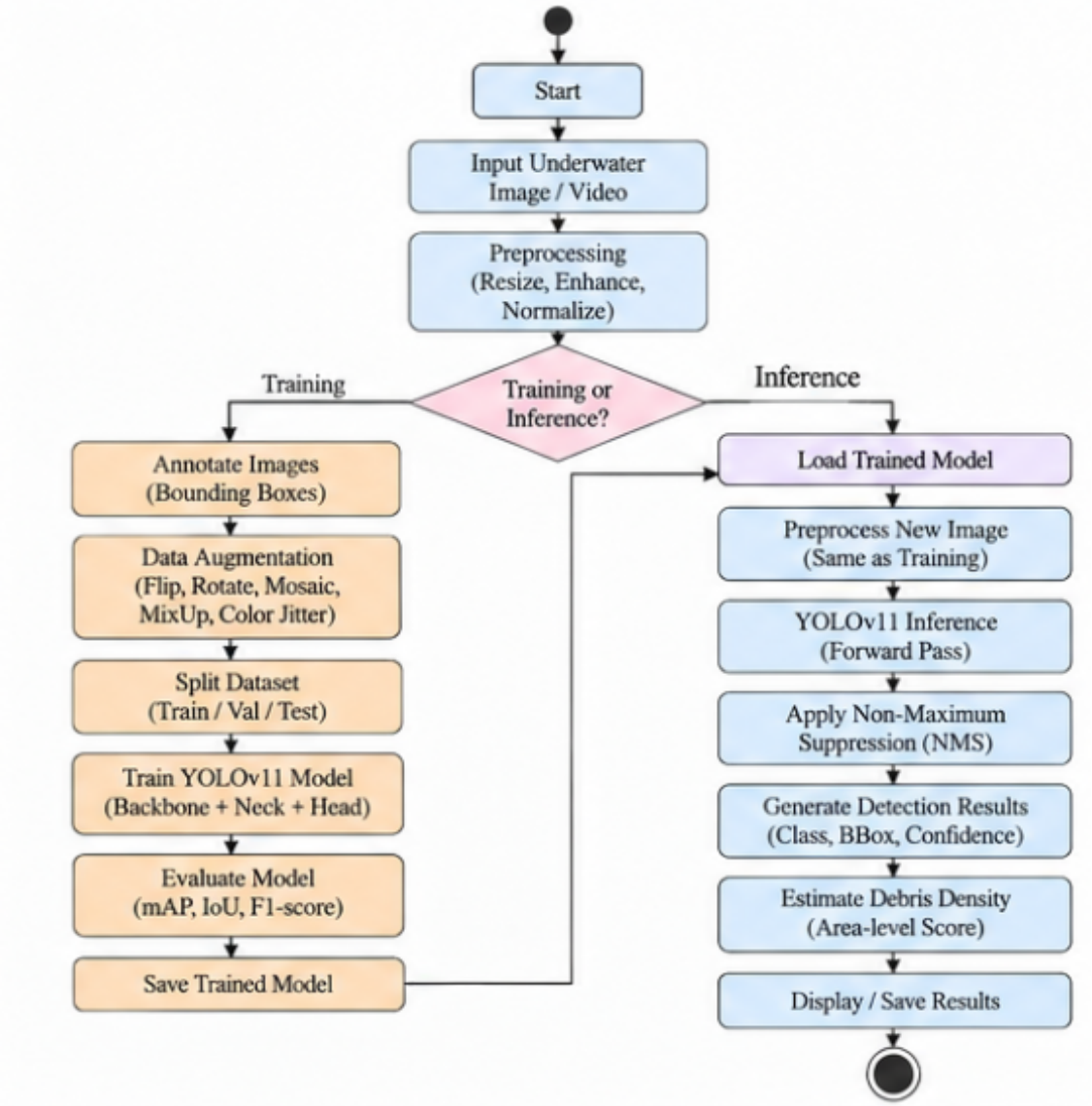


Figure 3.3: Activity Diagram of the Proposed NirmalSagar System

3.4.3 Use Case Diagram

The use case diagram illustrates the interaction between the users and the NirmalSagar system. It identifies the major functions provided by the system and the actors involved in those functions.

The primary user can upload underwater images or videos, view detection results, view debris density reports, and export the generated results. The system performs image preprocessing and marine debris detection using the trained YOLOv11 model. It also supports debris density estimation and generation of pollution-related reports.

The administrator or data manager is responsible for activities related to dataset management and annotation. The system also includes functions for model validation and evaluation. These

use cases provide an overall view of how different users interact with the proposed marine debris detection system.

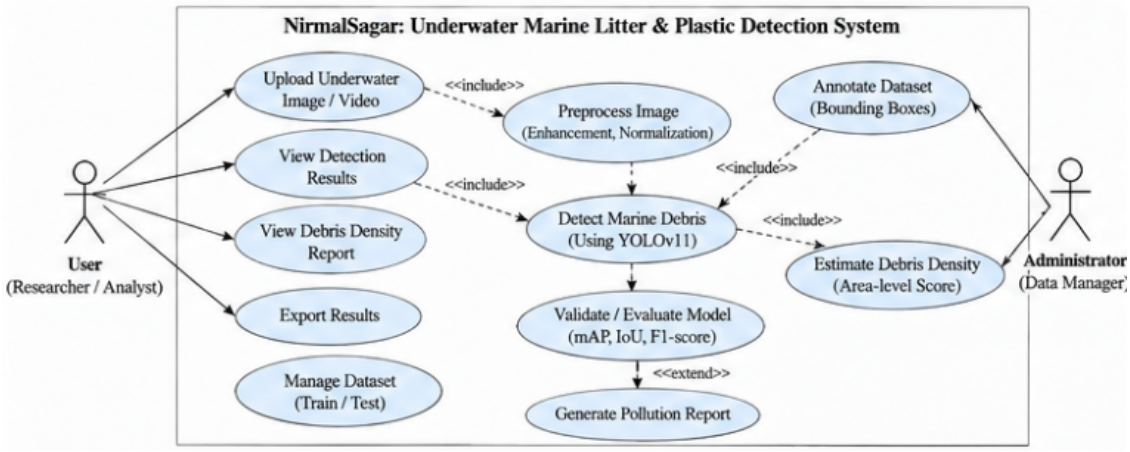


Figure 3.4: Use Case Diagram of the Proposed NirmalSagar System

3.4.4 Sequence Diagram

The sequence diagram represents the chronological interaction between the different components of the NirmalSagar system during the detection process. It shows how information is transferred from the user to the system and subsequently processed by the different modules.

The process starts when the user uploads an underwater image or video through the system interface. The input is passed to the preprocessing module, where the required preprocessing operations are performed. The processed input is then sent to the trained YOLOv11 model for inference.

After detection, the detection module processes the output and generates the detected marine debris information. The Debris Analyzer can then calculate the required debris density information. Finally, the result visualizer displays the detection results to the user, who can view or export the generated output.

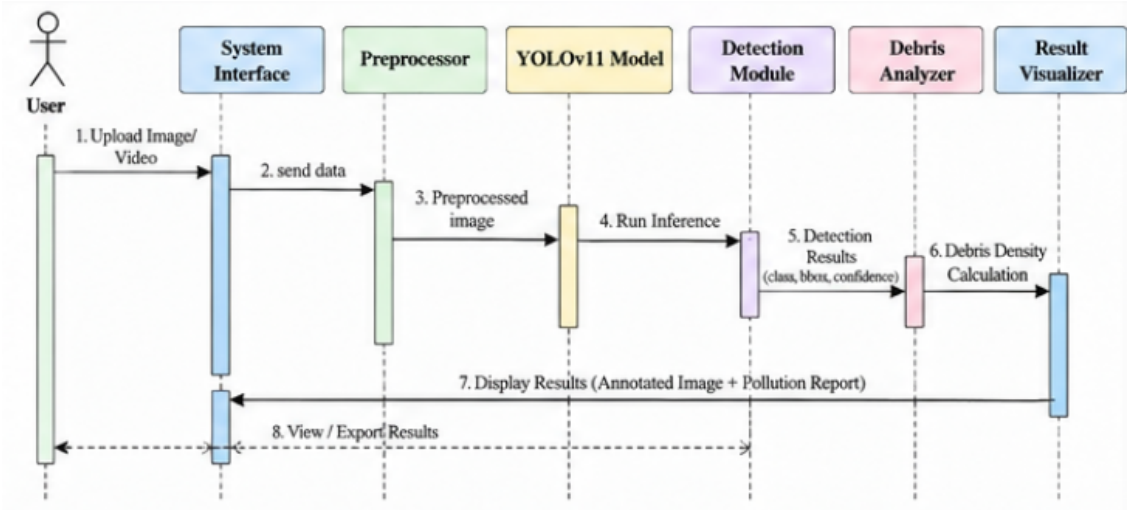


Figure 3.5: Sequence Diagram of the Proposed NirmalSagar System

Chapter 4

Project Implementation

4.1 Code Snippets

4.2 Steps to access the System

In Chapter 4 which is Project Implementation Project Groups should add at Max 6 snippets of your code or Screenshots which you feel are major & Technically critical along with brief description.

4.3 Timeline Sem VIII

In this section, Students need to show the timeline of their project milestone and how they have reached this stage of the project in graphical representation.

You can add more subsections as per project requirements.

Chapter 5

Testing

5.1 Software Testing

5.2 Functional Testing

As per outline in Chapter 5 Project Groups should jot down Which testing method out of Standard SW Testing methods is best suits for their Implementation. Write Theory in Breif about that Testing method as one paragraph. In second Para justify why it's best suits your project

Chapter 6

Result and Discussions

This shall form the penultimate chapter of the report and shall include a thorough evaluation of the investigation carried out and bring out the contributions from the study. The discussion shall logically lead to inferences and conclusions as well as scope for possible further future work.

Parameters	**Wormhole Attack**	**AODV**
Total No.of Sent Packets	85068	85068
Total No.of Received Packets	16252	68068
Total No.of Drop Packets	61948	16864
Throughput	98.01	410.24
Delay	0.0244	0.1528
Jitter	23.93	100.16

Table 6.1: Wormhole Attack Comparison with AODV Protocol

Chapter 7

Conclusion

The NirmalSagar project demonstrates the potential of artificial intelligence, deep learning, and computer vision for automated detection and localization of underwater marine debris. The project addresses an important environmental problem in which marine litter such as plastics, bottles, fishing nets, and other submerged waste can negatively affect marine ecosystems and marine life.

The proposed system focuses on the challenges associated with underwater image analysis. Factors such as turbidity, low visibility, low light, lighting variations, and complex underwater backgrounds can make marine debris difficult to identify. The project therefore considers underwater image enhancement and YOLO-based object detection as important components of the detection pipeline.

Another important aspect of the project is the problem of domain shift between different underwater datasets and environments. A model trained using images from one dataset or location may not perform consistently when applied to another environment with different visual characteristics. To address this challenge, the project considers domain adaptation and generalisation strategies such as synthetic augmentation, style transfer, and adversarial domain adaptation.

The project also considers evaluation across heterogeneous underwater marine-debris datasets including TrashCan 1.0, the Underwater Marine Debris in Shallow Waters dataset, and the JAMSTEC Deep-sea Debris Database. Standard evaluation measures such as mean Average Precision (mAP), Intersection over Union (IoU), and F1-score can be used to analyse detection performance and quantify the effect of domain shift.

In addition to individual object detection, NirmalSagar considers debris-density estimation by aggregating detections obtained from multiple frames. This provides an area-level indication of the concentration of marine debris and can support the prioritisation of areas for cleanup and marine conservation activities.

Overall, the proposed NirmalSagar system provides a computer-vision-based approach for assisting underwater pollution monitoring. By combining image processing, deep learning-based object detection, domain generalisation, and debris-density estimation, the project aims to reduce dependence on completely manual inspection and support more efficient identification of underwater marine litter.

Chapter 8

Future Scope

Although the proposed NirmalSagar system provides a foundation for automated underwater marine debris detection, several improvements can be considered to increase its robustness, accuracy, and practical applicability. One important area for future development is domain generalisation, since underwater images vary due to turbidity, illumination, camera characteristics, and geographical location. Future work can investigate advanced domain adaptation and generalisation techniques to improve performance on unseen and murkier Indian coastal and seafloor environments. The detection pipeline can also be improved by incorporating advanced underwater image enhancement techniques for better colour, contrast, illumination, and visibility restoration.

Future work can also focus on expanding the training and evaluation datasets with underwater images from different geographical locations, cameras, depths, and environmental conditions. This can improve the detection of small, rare, and partially occluded debris objects. The debris-density estimation module can further be enhanced by considering detection confidence, spatial distribution, and observations across multiple frames to provide more meaningful area-level pollution scores.

Another possible direction is real-time underwater monitoring through integration with underwater cameras, remotely operated vehicles (ROVs), or autonomous underwater vehicles (AUVs). Such integration can enable automated collection and analysis of underwater imagery. A user-friendly monitoring interface can also be developed to display annotated images, detection statistics, and debris-density information for analysis and decision-making.

In the long term, NirmalSagar can be developed into a comprehensive marine pollution monitoring framework by combining underwater image enhancement, robust object detection, domain generalisation, multi-frame analysis, and debris-density estimation. These improvements can support faster marine debris assessment and contribute to conservation and cleanup activities along Indian coastlines.

The system can also be extended to support continuous monitoring and comparison of marine debris levels over time. Historical detection data can be stored and analysed to identify changes in debris concentration across different locations and monitoring periods. Such information can help researchers and conservation teams understand pollution patterns and plan future monitoring and cleanup activities more effectively.

Another area of future development is optimising the detection model for practical deployment on resource-constrained devices. The model can be improved for faster inference, lower memory usage, and better computational efficiency while maintaining reliable detection performance. This can make NirmalSagar more suitable for underwater cameras and robotic platforms with limited computational resources.

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Appendices

Detailed information, lengthy derivations, raw experimental observations etc. are to be presented in the separate appendices, which shall be numbered in Roman Capitals (e.g. "Appendix I"). Since reference can be drawn to published/unpublished literature in the appendices these should precede the "Literature Cited" section.

Appendix-A: NS2 Download and Installation

1. Download ns-allinone-2.35.tar.gz from http://sourceforge.net/projects/nsnam/

2. Place ns-allinone-2.35.tar in your desired directory; like /home/vishal.

3. Go to terminal and do as following commands
sudo apt-get update
sudo apt-get install automake autoconf libxmu-dev build-essential

4. Extract ns-allinone-2.35 and after extracting go to folder ns-allinone-2.35 from Terminal as
$cd ns-allinone-2.35
$./install

5. Path Setting
$ gedit .bashrc

This command will open an existing file in editor. Just put the following path which is given bellow. [Remember that our ns-allinone path is /home/vishal. we will change this path according to our ns-allinone folder's path]

export PATH=$PATH:/home/vishal/ns-allinone-2.35/bin:/home/vishal/ns-allinone-2.35/tcl8.5.10/unix/home/vishal/ns-allinone-2.35/tk8.5.10/unix

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/home/vishal/ns-allinone- 2.35/otcl-1.14:/home/vishal/ns-allinone-2.35/lib

export TCL_LIRARY_PATH=$TCL_LIBRARY_PATH:/home/vishal/ns-allinone-2.35/tcl8.5.10/library

After this save and exit.

6. Now type in terminal to check that, is all command we entered in .bashrc is correct or not? And

To take the effect immediately
$source .bashrc

7. Then perform the validation test using this command.
$ ./validate

8. Run ns2 using this command
$ns

We will get % prompt in our terminal. Now ns2 has been installed.

Publication

Paper entitled **"Paper Title"** is presented at **"International Conference/Journal Name"** by **"Author Name".**

NOTE:

In Project Report Project Group will have to elaborate sections included as per work done after Sem VII. It can't be same Sem VII. Report must have proper description, Elaboration of Digrams included in Design & Implementation. It can't be just Digram on Page. You will have to write there why this has been included, What it is representing. Same is Applicable to snippets of Project implementation you have included in Report. Include your publications in your Report if You have.

**this note should be removed from your report